Double Integrals

Section 1: Fubini's Theorem over Rectangles

1 Evaluate $\int_0^2 \int_0^3 2xy dy dx$.

a. 18

b. 9

c. 36

d. 12

2 What does Fubini's Theorem state for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. The iterated integrals in either order are equal: $\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$

b. The double integral is always equal to $f(b, d) - f(a, c)$

c. The integral is zero if f is odd in x

d. The integral must be evaluated with x first

3 Evaluate $\iint_R (x + y) dA$ over $R = [0, 1] \times [0, 2]$.

a. 3

b. 2

c. 4

d. 1

Section 2: Double Integrals over General Regions

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4 Set up the double integral of $f(x, y)$ over the triangular region bounded by $y = 0$, $y = 2x$, and $x = 3$ as a Type I region.

a. $\int_0^3 \int_0^{2x} f(x, y) dy dx$

b. $\int_0^6 \int_0^3 f(x, y) dx dy$

c. $\int_0^{2x} \int_0^3 f(x, y) dx dy$

d. $\int_0^3 \int_{2x}^6 f(x, y) dy dx$

5 Evaluate $\int_0^1 \int_0^x 6xy dy dx$.

a. $\frac{3}{4}$

b. 1

c. $\frac{1}{2}$

d. $\frac{3}{2}$

6 What geometric quantity does $\iint_D 1 dA$ compute?

a. The area of the planar region D

b. The perimeter of region D

c. The volume under the paraboloid

d. The centroid

Section 3: Changing the Order of Integration

7 To evaluate $\int_0^1 \int_y^1 e^{x^2} dx dy$, which cannot be integrated directly, reverse the order of integration.

a. $\int_0^1 \int_0^x e^{x^2} dy dx$

b. $\int_0^1 \int_x^1 e^{x^2} dy dx$

c. $\int_0^1 \int_0^1 e^{x^2} dx dy$

d. $\int_y^1 \int_0^1 e^{x^2} dy dx$

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8 Evaluate the reversed integral $\int_0^1 \int_0^x e^{x^2} dy dx$.

a. $\frac{e-1}{2}$

b. $e-1$

c. $\frac{e}{2}$

d. 1

9 Reverse the order of integration for $\int_0^4 \int_{\sqrt{x}}^2 f(x,y) dy dx$.

a. $\int_0^2 \int_0^{y^2} f(x,y) dx dy$

b. $\int_0^2 \int_{y^2}^4 f(x,y) dx dy$

c. $\int_0^4 \int_0^{y^2} f(x,y) dx dy$

d. $\int_{\sqrt{x}}^2 \int_0^4 f(x,y) dx dy$

Section 4: Double Integrals in Polar Coordinates

10 What is the differential area element dA in polar coordinates?

a. $dA = r dr d\theta$

b. $dA = dr d\theta$

c. $dA = r^2 dr d\theta$

d. $dA = \frac{1}{r} dr d\theta$

11 Evaluate $\iint_D e^{-(x^2+y^2)} dA$ where D is the disk $x^2 + y^2 \leq 4$.

a. $\pi(1 - e^{-4})$

b. $2\pi(1 - e^{-4})$

c. πe^{-4}

d. $\frac{\pi}{2}(1 - e^{-4})$

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12 Convert $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)dydx$ to polar coordinates.

a. $\int_0^{\pi/2} \int_0^1 r^3 drd\theta$

b. $\int_0^{2\pi} \int_0^1 r^2 drd\theta$

c. $\int_0^{\pi/2} \int_0^1 r^2 drd\theta$

d. $\int_0^1 \int_0^1 r^3 drd\theta$

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